

Proof by Induction and Contradiction

Induction and Contradiction

Proof by Induction

The Template

To prove $P(n)$ holds for all integers $n \geq 1$:

1. **Base case.** Prove $P(1)$ directly.
2. **Inductive step.** Assume $P(k)$ holds for some $k \geq 1$ (the **induction hypothesis**). Use it to prove $P(k + 1)$.

Like dominoes: knock the first one over, and each one knocks over the next.

Example 1: Sum of the First n Integers

Claim

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

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Inductive step. Assume $1 + \cdots + k = \frac{k(k+1)}{2}$. Then

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1) \left(\frac{k}{2} + 1 \right) = \frac{(k+1)(k+2)}{2}. \quad \square$$

Example 2: Sum of Squares

Claim

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Base case ($n = 1$): $1 = \frac{1 \cdot 2 \cdot 3}{6} = 1.$

Inductive step. Assume

$$1^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}.$$

We need to show the formula holds for $k + 1$.

Example 2: Sum of Squares (Inductive Step)

Adding $(k + 1)^2$ to both sides of the induction hypothesis:

$$\begin{aligned}1^2 + \cdots + k^2 + (k + 1)^2 &= \frac{k(k + 1)(2k + 1)}{6} + (k + 1)^2 \\&= \frac{(k + 1)[k(2k + 1) + 6(k + 1)]}{6} \\&= \frac{(k + 1)(2k^2 + 7k + 6)}{6} \\&= \frac{(k + 1)(k + 2)(2k + 3)}{6}. \quad \square\end{aligned}$$

This matches the formula with $n = k + 1$.

Example 3: Geometric Series

Claim

For $r \neq 1$ and $n \geq 0$:

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}.$$

Base case ($n = 0$): $1 = \frac{r^1 - 1}{r - 1} = 1.$

Inductive step. Assume $1 + r + \cdots + r^k = \frac{r^{k+1} - 1}{r - 1}$. We need to show the formula holds for $k + 1$.

Example 3: Geometric Series (Inductive Step)

Adding r^{k+1} to both sides of the induction hypothesis:

$$\begin{aligned}1 + r + \cdots + r^k + r^{k+1} &= \frac{r^{k+1} - 1}{r - 1} + r^{k+1} \\&= \frac{r^{k+1} - 1 + r^{k+1}(r - 1)}{r - 1} \\&= \frac{r^{k+1} - 1 + r^{k+2} - r^{k+1}}{r - 1} \\&= \frac{r^{k+2} - 1}{r - 1}. \quad \square\end{aligned}$$

This matches the formula with $n = k + 1$.

A Useful Lemma

Fibonacci Addition Identity

For all $m \geq 1$ and $n \geq 1$:

$$F_{m+n} = F_{m-1} F_n + F_m F_{n+1}.$$

Proof (by induction on n , fixing m).

Base case ($n = 1$):

$$F_{m-1} F_1 + F_m F_2 = F_{m-1} \cdot 1 + F_m \cdot 1 = F_{m+1}.$$

Base case ($n = 2$):

$$F_{m-1} F_2 + F_m F_3 = F_{m-1} + 2F_m = (F_{m-1} + F_m) + F_m = F_{m+1} + F_m = F_{m+2}.$$

A Useful Lemma (Inductive Step)

Inductive hypothesis. Assume the identity holds for $n = k - 1$ and $n = k$.

Goal: show it for $n = k + 1$.

Using the Fibonacci recurrence $F_{m+k+1} = F_{m+k} + F_{m+k-1}$ and both hypotheses:

$$\begin{aligned} F_{m+k+1} &= F_{m+k} + F_{m+k-1} \\ &= [F_{m-1}F_k + F_mF_{k+1}] + [F_{m-1}F_{k-1} + F_mF_k] \\ &= F_{m-1} \underbrace{(F_k + F_{k-1})}_{F_{k+1}} + F_m \underbrace{(F_{k+1} + F_k)}_{F_{k+2}} \\ &= F_{m-1} F_{k+1} + F_m F_{k+2}. \quad \square \end{aligned}$$

Example 4: Every 5th Fibonacci Number is Divisible by 5

Recall $F_1 = 1$, $F_2 = 1$, $F_n = F_{n-1} + F_{n-2}$.

Claim

$5 \mid F_{5n}$ for all $n \geq 1$.

Base case ($n = 1$): $F_5 = 5$, so $5 \mid F_5$.

Inductive step. Assume $5 \mid F_{5k}$. Apply the Fibonacci addition identity with $m = 5k$, $n = 5$:

$$F_{5k+5} = F_{5k-1} F_5 + F_{5k} F_6 = 5 F_{5k-1} + 8 F_{5k}.$$

Both terms are divisible by 5:

- $5 F_{5k-1}$: contains the factor 5.
- $8 F_{5k}$: divisible by 5 by the induction hypothesis.

Therefore $5 \mid F_{5(k+1)}$. $\square$

Example 5: Edges in a Tree

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Inductive step. Every tree with $k + 1 \geq 2$ nodes has at least one **leaf** (a node of degree 1). Remove that leaf and its edge.

The remaining graph is a tree on k nodes, which by the induction hypothesis has $k - 1$ edges. Restoring the leaf adds back exactly 1 edge, giving

$$(k - 1) + 1 = k = (k + 1) - 1 \text{ edges. } \square$$

Proof by Contradiction

The Template

To prove statement P :

1. **Assume** $\neg P$ (the negation of P).
2. **Derive** a statement that is **always false** (a contradiction).
3. **Conclude** P must be true.

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Then $n = 2k + 1$ for some integer k , so

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. This contradicts n^2 being even. $\square$

Example 2: $\sqrt{2}$ is Irrational

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Proof. Assume $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$.

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Proof. Assume $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$.

Then $p^2 = 2q^2$, so p^2 is even $\Rightarrow p$ is even (Example 1). Write $p = 2m$:

$$4m^2 = 2q^2 \implies q^2 = 2m^2,$$

so q^2 is even $\Rightarrow q$ is even.

But then $2 \mid p$ and $2 \mid q$, contradicting $\gcd(p, q) = 1$. $\square$

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Let $N = p_1 p_2 \cdots p_k + 1$.

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Proof. Assume for contradiction that there are only finitely many primes: $p_1, p_2, \dots, p_k$.

Let $N = p_1 p_2 \cdots p_k + 1$.

Since $N > 1$, it has a prime factor p . That prime must be one of $p_1, \dots, p_k$, so $p \mid p_1 p_2 \cdots p_k$.

But $p \mid N$ and $p \mid p_1 \cdots p_k$ together imply

$$p \mid (N - p_1 p_2 \cdots p_k) = 1,$$

which is impossible. $\square$